

The effect of Surface Traps on the Interfacial Charge Dynamics in Layered Dielectrics

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Abstract– Space charge is a matter of concern in HVDC applications, causing considerable stress enhancement and consequent degradation. Apart from space charge within the volume of the insulation, interfacial charge at a multi-dielectric interface further aggravates the stress. The difference in permittivity and conductivity of the different dielectrics used in a composite insulation system are primarily responsible for interfacial charge. However, surface conditions and trap characteristics at the dielectric-dielectric interface have a significant role to play, and neglecting these effects would lead to erroneous estimation of interfacial charge, and poor insulation design. Hence, an improved charge transport model is proposed in this paper to model the charge dynamics at dielectric-dielectric interfaces. The effect of surface traps on a 300 μ m 2-layer LDPE is studied through space charge data obtained from Pulsed Electro Acoustic (PEA) measurements.

Keywords— Bipolar Charge Transport, Space Charge, LDPE, Interfacial Charge

I. INTRODUCTION

Sustained DC field application is responsible for space charge accumulation in HVDC insulation systems. For such applications, it is common to use layered dielectrics to improve its dielectric withstand capability [1]. However, use of layers might encourage space charge accumulation in the material which can significantly distort the electric field distribution inside a material [2,3]. Surfaces at interfaces may have broken chain bonds, chain folds, impurities etc resulting in a large number of charge traps leading to interfacial charge accumulation [4]. Hence, it is important to study the interfacial charge dynamics of layered insulation systems.

The Maxwell-Wagner theory provides an estimate of the temporal variation of interfacial charge in layered dielectrics and is given by Equation (1) below [5]. The schematic of layered dielectrics is shown in Fig.1.

$$\rho(t) = \frac{V(\sigma_1 \varepsilon_2 - \sigma_2 \varepsilon_1)}{d_1 \sigma_2 + d_2 \sigma_1} (1 - e^{-t/\tau}) \quad \text{Eqn-1}$$

$$\tau = \frac{(d_1 \varepsilon_2 + d_2 \varepsilon_1)}{(d_1 \sigma_2 + d_2 \sigma_1)}$$

where, $\rho(t)$ is the interfacial charge density in C/m², V is the applied voltage, while, ε , σ and d are the permittivity, conductivity and thickness respectively.

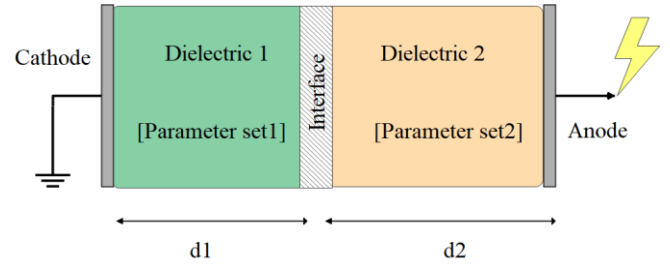


Figure 1. layered dielectrics

The material parameter set for the two dielectrics are distinguished by appropriate subscripts 1 and 2.

Das and Gupta studied charge accumulation at an epoxy-epoxy interface. Since the layers are of the same material, ideally no interfacial charge should be present. However significant interfacial charge accumulation was observed and was attributed to surface states [4]. More specifically, negative interfacial charge was observed. This phenomenon cannot be explained by the Maxwell-Wagner theory and is due to surface traps. Therefore, we propose an improved Bipolar Charge Transport (BCT) model for multi-dielectrics (layered) in order to model the interface trapping phenomena. In this paper, the interface charge trapping in a layered LDPE is modelled. Additionally, the role of charge trapping frequency and trap depth at the interface of a 300 μ m LDPE-LDPE layered dielectric is investigated using the BCT multi-dielectric model in conjunction with space charge results from Pulsed Electro Acoustic (PEA) measurements.

II. THE BCT MULTI-DIELECTRIC MODEL

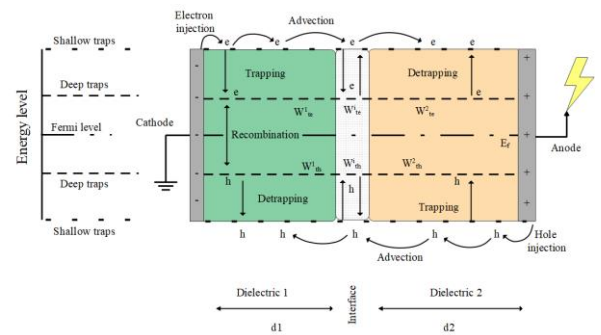


Figure 2. BCT for Multi-Layer dielectrics

The charge dynamics in a dielectric is determined by charge advection as well as charge reaction processes. Charge

advection constitutes of drift, diffusion, injection and extraction of charges. Charge reaction constitutes of trapping, detrapping and recombination. This is shown schematically in Fig.2. The set of simultaneous differential equations governing BCT is reported in [6,7]. In the proposed model, interfacial charge accumulation is assumed to be the difference between inflow of charge carriers (holes and electrons) and outflow of charge carriers at the interface, and governed by (2).

$$\frac{\partial \rho_{\mu}^i(e,h)}{\partial t} = J_{(e,h)}^{i+} - J_{(e,h)}^{i-} + S_{\mu}^i(e,h) \quad (2)$$

where ρ_{μ} (C/m²) represents the mobile charge accumulation at the interface. J^{i+} and J^{i-} represents the inflow and outflow of charge carriers at the interface (units A/m²). S_{μ}^i represents the reaction terms for mobile carriers and is evaluated using (3)-(7).

$$S_{ue}^i = D_{e,e}^i \cdot \rho_{te} - T_{e,e}^i \cdot \rho_{\mu e}^i \cdot \left(1 - \frac{\rho_{te}^i}{eN_t}\right) \quad (3)$$

$$S_{uh}^i = D_{h,h}^i \cdot \rho_{th}^i - T_{h,h}^i \cdot \rho_{\mu h}^i \cdot \left(1 - \frac{\rho_{th}^i}{eN_t}\right) \quad (4)$$

$$\frac{\partial \rho_{te}^i}{\partial t} = T_{e,e}^i \cdot \rho_{\mu e}^i \cdot \left(1 - \frac{\rho_{te}^i}{eN_t}\right) - D_{e,e}^i \cdot \rho_{te} \quad (5)$$

$$\frac{\partial \rho_{th}^i}{\partial t} = T_{h,h}^i \cdot \rho_{\mu h}^i \cdot \left(1 - \frac{\rho_{th}^i}{eN_t}\right) - D_{h,h}^i \cdot \rho_{th}^i \quad (6)$$

$$D_{e,h}^i = v_{ate} \cdot \exp\left(-\frac{W_{e,h}^i}{k_B T}\right) \quad (7)$$

where, D^i is the detrapping frequency of charge carriers given by Equation (7). N_t is the number densities of traps in 1/m². k_B is the Boltzmann's constant and equals 1.38×10^{-23} J/K. T is the temperature and equal to 298K. v_{ate} is the attempt to escape frequency of charge carriers and equal to 6.2×10^{12} [7]. ρ_{μ}^i and ρ_t^i represent the mobile and trapped charge densities in C/m² at the interface. The net interface charge density is the sum of mobile and trapped charge densities at the interface (C/m²). Further, this charge is divided by the finite element thickness (h) to evaluate the interface volume charge density in C/m³. A single deep trap level is assumed at the interface for simplicity. The mobile electrons and holes that flow into the interface are trapped with charge trapping frequencies of $T_{e,e}^i$ and $T_{h,h}^i$ respectively. The trap depths at the interface are $W_{e,e}^i$ and $W_{h,h}^i$ for electrons and holes respectively. The quantities corresponding to electrons and holes are distinguished by appropriate subscripts of e and h throughout this paper. Further, trapped charges at the interface can thermally detrapp and become mobile with time, and then drift into the dielectrics depending on the direction of electric field. This process is included in the current model. The charge dynamics in the bulk is computed using the traditional BCT model [6]. Other interface reactions such as recombination, partial discharge etc. are ignored at present.

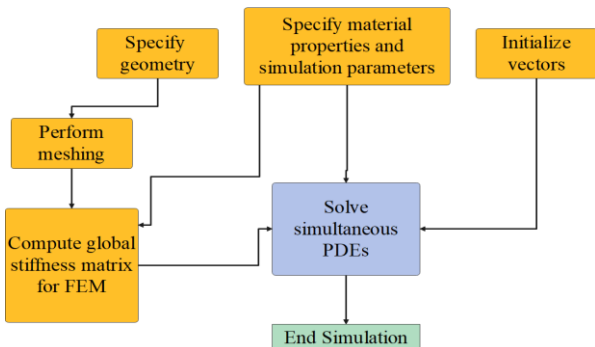


Figure 3. Simulation flow chart

The flow chart for modelling charge dynamics is given in Fig.3. The sample geometry viz. thickness and the material properties such as permittivity, trapping characteristics, injection barrier potentials etc. are input to the model. The problem is formulated in 1-dimension for the Finite Element Method (FEM) and Finite Volume Method (FVM) Computations respectively. FVM is used to evaluate the flow of electrons and holes in the dielectric [2]. The electric field is computed at all spatial locations along the sample thickness through the solution of Poisson's equation (8) using FEM.

$$\frac{\partial}{\partial x} \left(-\epsilon \frac{\partial \phi(x,t)}{\partial x} \right) = Q(x, t) \quad (8)$$

A static mesh is used. This is computationally efficient. The charge advection and reaction process in the bulk and the interface are simultaneously solved using MATLAB's implicit PDE solver.

III. EXPERIMENTAL PROCEDURE

Two layers of 170μm thick LDPE sheets are hot pressed at 60°C at a pressure of 3 bar for 3 hours. The overall thickness of the pressed sample is 300 μm. Circular samples of diameter 30mm are used. Aluminium electrodes of 27mm diameter are pasted on both sides of the sample for Pulsed Electro Acoustic (PEA) measurements. The PEA measurement system used is Tech.Imp Flat PEA module. The polarization time is set to 4300s and the applied electric field is 30kV/mm. The experimental results are discussed in the following section.

IV. RESULTS

Figure 4 shows the variation of space charge in the LDPE-LDPE layered dielectric for an applied electric field of 30kV/mm. Although both the layers are of the same material (LDPE), interfacial charge is observed. Negative charge quickly builds up at the interface within 200s of polarization. Over time, the interface charge slowly diminishes. At about 1500s, inflow of positive charges at the interface is observed. The wave of holes pass through the interface from one dielectric to another, while the electrons get trapped into the interface. The location of the interface is at 145μm and at about 150s the interfacial charge starts to build up. The space charge distribution at 150s is shown in Fig.5.

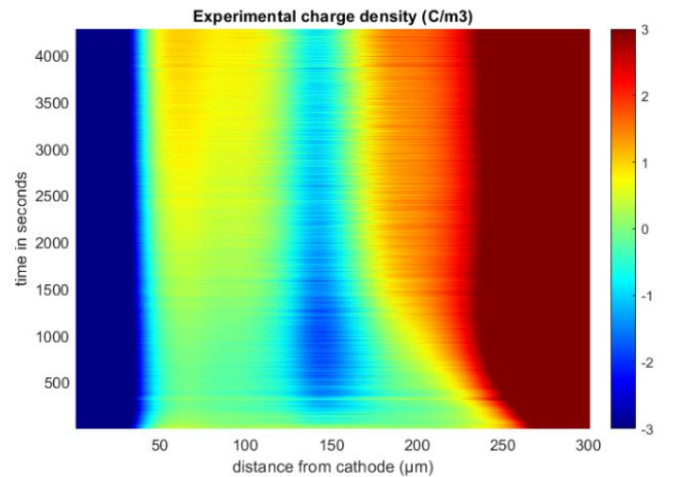


Figure 4. Experimental charge density (C/m³) for LDPE-LDPE layered dielectric

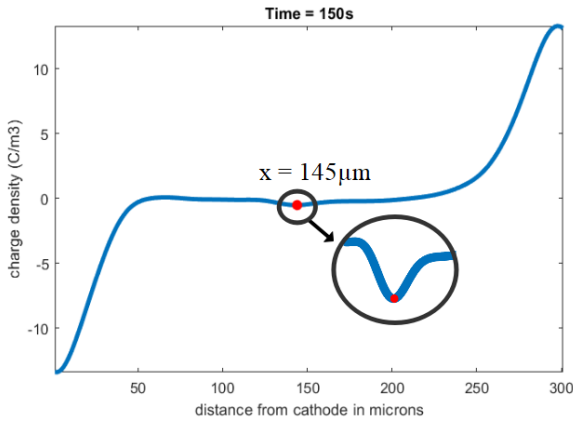


Figure 5. Experimental charge density (C/m³) at 150s of polarization

The BCT-Multidielectric model only predicts the behaviour of internal charges. The capacitive and image charges need to be accounted for separately. Finally, the estimated net charge distribution is passed through a Gauss filter to eliminate the abrupt spatial variations in computational charge density [7]. The equations pertaining to capacitive, image charges and the Gaussian function are given by (9) through (12) and are elaborated in [7]. The optimization process and corresponding results are discussed in the following sections.

$$q_{cap} = \frac{\epsilon_o \epsilon_r V}{d.h} \quad (9)$$

$$q_1 = \left(-\frac{1}{h}\right) \int_0^d \left(1 - \frac{x}{d}\right) \cdot q(x) \quad (10)$$

$$q_2 = \left(-\frac{1}{h}\right) \int_0^d \left(\frac{x}{d}\right) \cdot q(x) \quad (11)$$

$$g = \left(\frac{h}{s\sqrt{2\pi}}\right) \cdot \exp\left(-0.5 * \left(\frac{x}{s}\right)^2\right) \quad (12)$$

where, q_{cap} is the capacitive charge density, q_1 and q_2 are image charge densities at cathode and anode, g is the Gaussian function used for Gaussian filtering of the net computational space charge distributions.

Trust Region Algorithm (TRA) is used for (Least Squares) fitting the experimental space charge data with the computational space charge data. Equations pertaining to the Trust Region Method can be found in [7]. The optimization parameters considered are bulk charge trapping frequency, injection barrier potential, bulk mobility, bulk trap depth, interface charge trapping frequency, interface trap depth of holes and electrons respectively. Hence a 12 parameter optimization is carried out. Figures 6.a and 6.b show the experimentally-obtained space charge density and computationally-obtained space charge density respectively. A fairly good match is observed between the two.

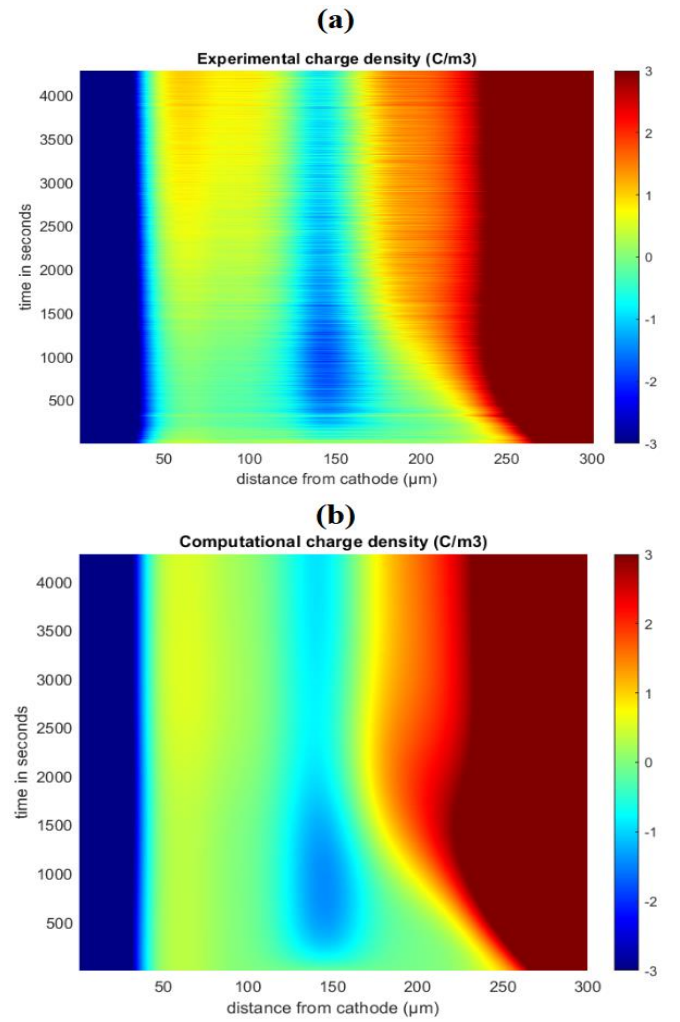


Figure 6. (a) Experimental charge density (C/m³) (b) Computational charge density (C/m³) for LDPE-LDPE layered dielectric

Fig.7 shows the variation of charge density at the interface ($x=145\mu m$) for both experimental and computational studies. The computational interfacial charge density profile matches the experimental interfacial profile reasonably well. The values of the optimal parameter set for which a good match is obtained are enumerated in Table I.

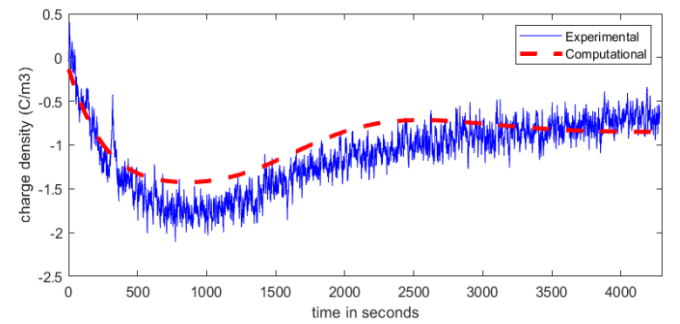


Figure 7. Experimental and computational interfacial charge profiles.

TABLE I
OPTIMAL BCT PARAMETER SET

Species Parameter	electrons	holes
Bulk trapping frequency (1/s)	0.0033	1.5101
Bulk charge mobility (m ² /Vs)	1050 x 10 ⁻¹⁵	203.67 x 10 ⁻¹⁵
Bulk trap depth (eV)	0.7923	0.8719
Injection barrier potential (eV)	1.1603	1.1097
Interface trapping frequency (1/s)	4.5010	0.2379
Interface trap depth (eV)	1.1301	0.7947

V. DISCUSSION

From Table I it is inferred that the bulk charge trapping frequency of holes is higher than that of electrons and the injection barrier potential of electrons is higher than that of holes. These results compare well with the reported literature [2,7]. Electron mobility is higher than hole mobility. The interface trapping frequency of electrons is much higher than that of holes. Higher the number density of the traps, higher is the charge trapping frequency. This implies that a surface has abundance of electron traps. This is responsible for the net negative charge accumulation at the interface seen in measurements. Also, the interface trap depth is much lower for holes than electrons. This implies that the electrons at the interface are trapped for a long duration due to a high trap depth. On the other hand, holes are less likely to be trapped at the interface (due to low trapping frequency). Also, trapped holes at the interface can easily escape the interface due to a low interface trap depth (0.79eV). Since, the interface trap depth of holes is lower, they migrate through the interface more easily than electrons.

VI. CONCLUSIONS

In this work, the effect of interface (surface) traps on layered LDPE-LDPE dielectric is studied through BCT multidielectric modelling and experimentally obtained space charge distributions. From the above study, the following points are concluded:

- In an LDPE-LDPE layered dielectric, negative interfacial charge is likely to occur and is attributed to surface states,
- the interface trap depth of electrons is higher than that of holes for a LDPE-LDPE layered dielectric,
- the interface trapping frequency of electrons is much higher than that of holes.

VII. ACKNOWLEDGMENT

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